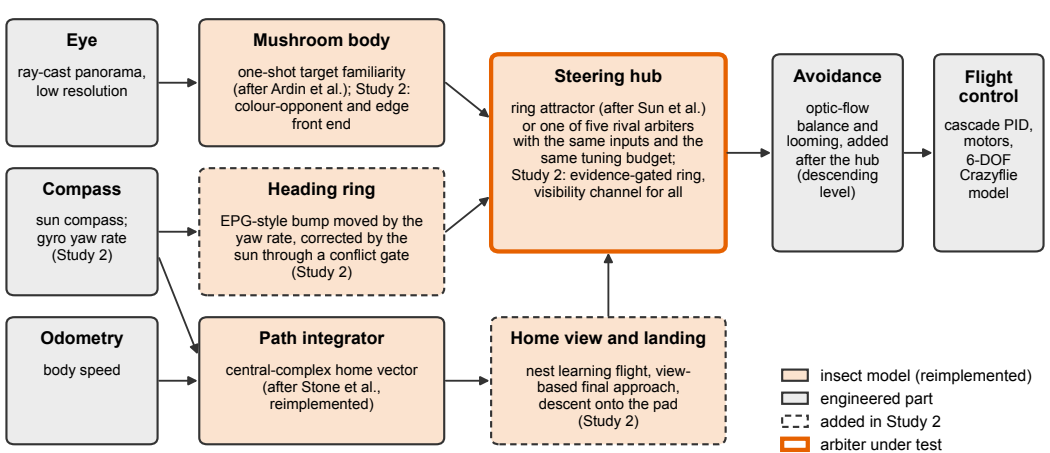
Eye
ray-cast panorama, low resolution
Mushroom body
one-shot target familiarity (after Ardin et al.); Study 2: colour-opponent and edge front end
Steering hub
ring attractor (after Sun et al.) or one of five rival arbiters with the same inputs and the same tuning budget; Study 2: evidence-gated ring, visibility channel for all
Avoidance
optic-flow balance and looming, added after the hub (descending level)
Flight control
cascade PID, motors, 6-DOF Crazyflie model
Compass
sun compass; gyro yaw rate (Study 2)
Heading ring
EPG-style bump moved by the yaw rate, corrected by the sun through a conflict gate (Study 2)
Odometry
body speed
Path integrator
central-complex home vector (after Stone et al., reimplemented)
Home view and landing
nest learning flight, view-based final approach, descent onto the pad (Study 2)
insect model (reimplemented)
engineered part
added in Study 2
arbiter under test